4. Functional Specification

1. Functional Description

This interface consists of sending maintenance order data to CMX, triggered by the occurrence of 4 specific situations:

- Case 1: Initial release of a maintenance order.
- Case 2: A maintenance order changes after it was released.
- Case 3: A maintenance order is closed (TECO) after it was released, so CMX does not process it.
- Case 4: A maintenance order is locked or unlocked after it was released, or after it was locked, respectively.
-

Only maintenance orders that meet each one of the following conditions are eligible to be sent to CMX:

- The order type of the maintenance order is “YA07”.
- The plant of the maintenance order is in scope for the search.
- The user status CMXC “CMX Confirmed” (internal ID = E0010) **is not** active for the maintenance order.
- The functional location to which the order is assigned must have the value “CMX” for the characteristic called “INTERFACE”.
- The order has never been sent before or has changed since the last time the program ran.*

Order type and plant should be set as parameters, to be easily changed directly in S/4HANA.

*If there is the possibility to transfer the orders real-time, without impacting the performance of the system while the user is saving the order, this is the preferred method. As an alternative, a small delay (transfer every 5-10 minutes) is also acceptable.

The program should allow re-execution, in case of errors during the transfer. To allow this flexibility, the “last time the program ran” should also be flexible, for example through the usage of a dynamic date, but manually changeable.

In addition, the interface will assign user status CMXS (technical ID = E0014) to any order that is sent to CMX in order to track which orders have been sent and which were not. This is also important in order to be able capture orders that were not eligible to be sent in

previous runs of the interface, but that will now be due to a change in the classification data of the technical object assigned to it.

A condition filtering operations by work center, which was implemented in the ECC system, will not be in S/4 HANA, as it is deemed to be redundant due to the order already having a separate order type for calibration, which was not available in the ECC system.

It is worth noting that it is possible that an order that may have initially not met some of the conditions listed above, after updating the technical object assigned to it, may now meet all such conditions. In the current system, the solution that was implemented consists of a user-exit that runs when an order is changed. However, if the change occurred in the master data after the order was saved/created, this will not be captured, and the order will incorrectly be missed or not sent to CMX. In order to address this issue, the new interface in S/4 HANA will be triggered by a background job that will run at regular intervals (or on demand) and which will evaluate all of the conditions described before and send the orders that meet them, independently of whether the order itself was updated or not.

Search Algorithm

The program will run with the following selection criteria:

Field	Mandatory	Data Element	Default	Format / Options
Order	No	AUFNR	None	Multiple selection
Plant	Yes	WERKS_D	None	Multiple selection
Order Type	Yes	AUART	None	Multiple selection
Check from Date	Yes	SY-DATUM	System date	Single value
Check from Time	Yes	SY-UTIME	00:00:00	Single value
Test run	No	-	<not selected>	-

Some of the conditions mentioned at the start of this section will be enforced as part of the selection variant used to run the program, such as the plant(s) and the date and time to check from (which will be set by default as the system date and time minus a fixed number of minutes). The algorithms to determine the orders to be sent will be executed in steps, as follows:

#	Step	Algorithm
1	Determine internal ID of the INTERFACE characteristic	Read field ATINN from table CABN where ATNAM = “INTERFACE”.
2	Determine list of released/technically complete orders	Read all entries from view VIAUFKST where: 1. AUART matches the selection criteria and… 2. IWERK matches the search criteria and… 3. IDAT1 is not blank or IDAT2 >= check from date (from selection criteria).
3	Determine object id of technical object for CMX relevancy validation	For each order found in step 2, do the following: 1. Read the value of field VIAUFKST-EQUNR and then: a. If not blank, it means the order was created for an equipment. In this case, take now the value of field VIAUFKST-EQUNR. b. If blank, it means the order was created only for a functional location. In this case, take the value of field VIAUFKST-TPLNR. 2. Read the value of CUOBJ from table INOB where: a. OBJEK = the value obtained in step 3.1.
4	Determine validity of each order	For each order found in step 2, do the following: 1. Read table JEST where: a. OBJNR = value of field VIAUFKST-OBJNR found in step 1 and… b. STAT = “I0002” or “I0045” and… c. INACT is blank. 2. Then depending on the outcome: a. If a match is found, the order is valid. Continue to step 4.3. b. If no match is found, discard order, continue with the next and go back to step 4.1. 3. Read table JEST where: a. OBJNR = value of field VIAUFKST-OBJNR found in step 1 and… b. STAT = “E0010” (CMX confirmed). 4. Then depending on the outcome:

		a. If a match is found, check the value of JEST-INACT and then: i. If blank, discard the order, continue with the next and go back to step 4.1. ii. If “X”, the order is valid. Continue to step 4.5. b. If no match is found, the order is valid. Continue to step 4.5. 5. Read table AUSP where: a. OBJEK = value of field INOB-CUOBJ obtained in step 3.2 and… b. ATINN = CABN-ATINN (found in step 1) and… c. ATWRT = “CMX”. 6. Then depending on the outcome: a. If a match is found, it means the order is valid and can now be checked further. b. If no match is found, discard the order, continue with the next and go back to step 4.1.
5	Determine whether the order is new or has been changed since the date and time specified in the selection criteria (check from date and time)	For each order that passed the verification in step 4: 1. Concatenate the following values into one text string: SY-MANDT + “30” + the order number (with leading zeros). 2. Read the maximum combined value of fields UDATE and UTIME from table CDHDR where: a. OBJECTCLAS = “ORDER” and… b. OBJECTID = concatenated string from step 5.1. 3. Compare them with the combined values of date and time that were specified as check from date and time in the selection criteria then: a. If the values from table CDHDR are greater, the order must be sent. b. If they are lesser, the order was not changed since the date and time in the selection criteria, but in this case, we still have one more check to do. Go to step 5.4. 4. Read table JEST where: a. OBJNR = value of field VIAUFKST-OBJNR and…

		b. STAT = “E0014” (sent to CMX) and… c. INACT is blank. 5. Then depending on the outcome: a. If no match is found, it means the order has never been sent before, most likely due to an error in master data. b. If an entry is found, it means the order has already been sent and must not be sent again.
6	Determine language dependent (English) names of the system statuses	Carry out the following, depending on the specific status: 1. Released: Read field TXT04 from table TJ02T where ISTAT = “I0002” and SPRAS = “EN”. 2. Locked: Read field TXT04 from table TJ02T where ISTAT = “I0043” and SPRAS = “EN”. 3. Technical completion: Read field TXT04 from table TJ02T where ISTAT = “I0045” and SPRAS = “EN”.
7	Add operation number	For each order that passed all the checks, take the first active operation: Read fields VORNR, AUFPL and APLZL from table VIAUFK_AFVC where AUFNR = the order number sorting by APLZL. Then for each match: Read field OBJNR from table AFVC where: AUFPL = VIAUFK_AFVC-AUFPL and APLZL = VIAUFK_AFVC-APLZL. Read table JEST where: OBJNR = AFVC-OBJNR and STAT = “I0013” (deletion indicator) and INACT is blank. Then depending on the outcome: If a match is found, it means the operation was deleted. In this case, discard this entry and continue to the next entry in AFVC. If a match is not found, it means the operation is still valid. In this case, exit the loop and go back to step 7.1 to process the next order.

In regards to the selection parameter called “Test run”:

- If its value is blank, then the date and time of execution of the program will be logged.
- It its value is “X”, then no date or time of execution of the program will be logged.

If the program ran in live mode (Test Run = blank), then the following function module must be executed to assign the status “Sent to CMX”:

Function Module = BAPI_ALM_ORDER_MAINTAIN

Parameters:

- IT_HEADER:
  - ORDERID = Order number (VIAUFKST-AUFNR).
- IT_USERSTATUS:
  - USER_ST_TEXT = “CMXS”.
  - LANGU = “EN”.
  - LANGU_ISO = “E”.

Program Output

Once the program run has finished, there will be an output displaying all the orders that were sent. The data to be displayed will be:

- Message type for CMX = CREATE.
- Plant (VIAUFKST-WERKS).
- Order number (VIAUFKST-AUFNR).
- Operation number (VIAUFKST-AFVC-VORNR).
- Order header status (TJ02T-TXT04).
- Technical object:
  - If VIAUFKST-EQUNR is not blank, then VIAUFKST-EQUNR.
  - If VIAUFKST-EQUNR is blank, then ILOA-TPLNR.
- Basic start date (VIAUFKST-GSTRP).
- Basic finish date (VIAUFKST-GLTRP).
- Final due date (VIAUFKST-LACD_DATE).
- Error text. To be displayed only if the function module to update the user status failed. In this case, it should show the error message 72 character text.

However, depending on the case in which each order is, the values of those columns will be displayed when they need to be sent to CMX only. To see what values are displayed (and sent) depending on the case, see the following part of this document, where all scenarios are described.

Finally, regardless of how the data is displayed, and unless the program was run in test mode, all orders listed in the output will be sent to CMX.

Possible Scenarios

Case 1: Order Initial Released

This occurs when an order is created and then released, either immediately upon creation or at a later date. In this case, the following data must be sent:

- Message type for CMX = CREATE.
- Order number (VIAUFKST-AUFNR).
- Operation number (VIAUFK_AFVC-VORNR).
- Order header status (TJ02T-TXT04) with the value for “Released” status.
- Technical object:(see logic described in subsection “Program Output”).
- Basic start date (VIAUFKST-GSTRP).
- Basic finish date (VIAUFKST-GLTRP).
- Final due date (VIAUFKST-LACD_DATE).
- System ID (T30,Q30,P10,Q10...).

Case 2: Order Changed After Initial Release

This occurs when an order is updated after it was originally released and thus it must be sent again (if it was already sent) to CMX. In this case, the following data must be sent:

- Message type for CMX = CREATE.
- Order number (VIAUFKST-AUFNR).
- Operation number (VIAUFK_AFVC-VORNR).
- Order header status (TJ02T-TXT04) with the value for “Released” status.
- Technical object: (see logic described in subsection “Program Output”).
- Basic start date (VIAUFKST-GSTRP).

- Basic finish date (VIAUFKST-GLTRP).
- System ID (T30,Q30,P10,Q10...).

Case 3: Order Technically Completed After Initial Release

This occurs when an order is technically completed after being released. Typically, this will happen if the order was not meant to be sent to CMX, and then it was completed technically only to inform CMX that it will not be executed and must be closed in it. In this case, the following data must be sent:

- Message type for CMX = CREATE.
- Order number (VIAUFKST-AUFNR).
- Operation number (VIAUFK_AFVC-VORNR).
- Order header status (TJ02T-TXT04) with the value for "Technical Completion" status.
- Technical object: See logic described in subsection "Program Output".
- System ID (T30,Q30,P10,Q10...).

Case 4: Order Locked/Unlocked After Initial Release

This occurs when an order is either locked, which must be communicated to CMX, or unlocked after it was already communicated to CMX that it had previously been locked. In this case, the following data must be sent:

- Message type for CMX = CREATE.
- Order number (VIAUFKST-AUFNR).
- Operation number (VIAUFK_AFVC-VORNR).
- Order header status: Read table JEST where OBJNR = VIAUFKST-OBJNR and STAT = "I0043". If a record is found that meets these conditions, then, depending on the value of field JEST-INACT:
  - If it is blank, then send the value of field TJ02T-TXT04 for the "Locked" status.
  - If it is "X", then send the value of field TJ02T-TXT04 for the "Released" status.
- Technical object: See logic described in sub-section "Program Output".
- System ID (T30,Q30,P10,Q10...).

Field Mapping

S. No.	Description	SAP Backend Table	SAP Field	SAP API/Entity	SAP Odata Field	Logic
1	Order Message Status	-	‘CREATE‘	I_MaintenanceOrderDEX		Fixed value: “CREATE”
2	Order Number	VIAUFKST	AUFNR	I_MaintenanceOrderDEX	MaintenanceOrder	_Order
3	Operation	VIAUFK_AFVC	VORNR	I_MaintenanceOrderDEX	MaintenanceOrderOperation	_MaintenanceOrderOperation
4	Order Header Status	TJ02T	TXT04	I_MaintenanceOrderDEX	SystemStatusText	Values are dependent on the case for each order: Case 1: REL Case 2: REL Case 3: TECO Case 4: LKD if order is locked and REL if order is unlocked (see algorithm)
5	Technical Object	VIAUFKST Or ILOA	EQUNR Or TPLNR	I_MaintenanceOrderDEX I_FunctionalLocation I_FunctionalLocationLabel I_Equipment	FunctionalLocationLabelName	First, check if the order is assigned to an equipment, then: If assigned to an equipment, use that value here. If not, read the functional location as follows: From I_FunctionalLocationLabel use: FunctionalLocationLabelName
6	Basic Start Date	VIAUFKST	GSTRP	I_MaintenanceOrderDEX	MaintOrdBasicStartDate	Get MaintOrdBasicStartDate
7	Basic End Date	VIAUFKST	GLTRP	I_MaintenanceOrderDEX	MaintOrdBasicEndDate	Get MaintOrdBasicEndDate

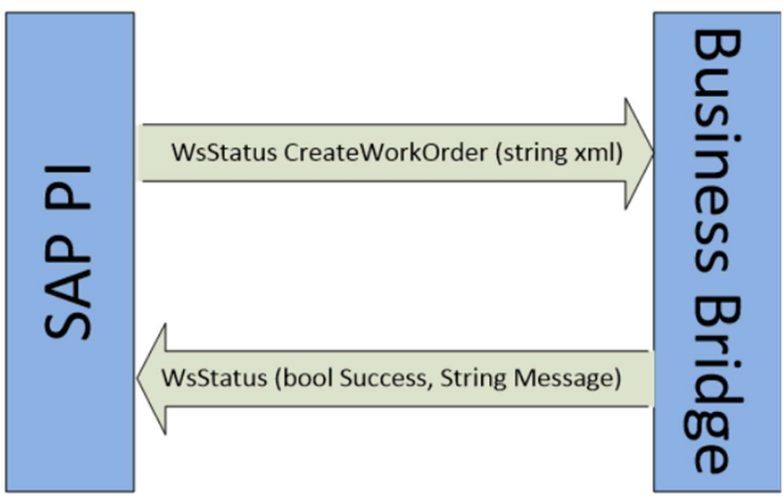
8	Final Due Date	VIAUFKST	LACD_DATE	I_MaintenanceOrderDEX	LatestAcceptableCompletionDate	Get LatestAcceptableCompletionDate

Communication

The Interfaces are realized following the Documentation “Business Bridge Configuration” provided by Beamex.

Technically there are 3 Interfaces covering all communication between SAP and CMX.

For IF41S4-01 the following communication takes place:



Unit Testing

Case 1a: Initial release (test mode)

- Select 5 functional locations of which:
  - 4 will have the value “CMX” for characteristic INTERFACE.
  - 1 will not.
- Create a maintenance order (IW31) for each functional location with the following data:
  - Order type = YA07Basic start date = today.
  - Basic end date = any.
- Release the orders (IW32).
- Run the program with the following selection criteria:
  - Plant = ZXZZ.
  - Order type = YA07.
  - Last execution date = today.
  - Last execution time = 00:00:00.
  - Test run = “X”.
- The program output will display the 4 orders with all required columns.

Case 1b: Initial release (live mode)

- Run the program with the same selection criteria, except now the test run must be unselected.
- The program output will display the 4 orders with all required columns. Note the exact time when the program was run.

Case 2a: Change After Initial Release (test mode)

- Select 2 random orders of the 3 that were sent before.
- Change the basic end date in one of them and the work center in the other (IW32).
- Run the program with the same selection criteria as in the previous test, but in test mode.
- The program output will display the 2 orders with all required columns for case 2.

Case 2b: Change After Initial Release (live mode)

- Run the program with the same selection criteria, except now the test run must be unselected.
- The program output will display the 2 orders with all required columns for case 2. Note the exact time when the program was run.

Case 3: Cancellation After Initial Release

- Open one of the 2 orders shown in the previous test and close it from the menu: Order > Functions > Complete > Complete (Technically) and save order (IW32).
- Run the program with the same selection criteria (test run unselected).
- The program output will display the order with all required columns for case 3 (status = TECO).

Case 4a: Lock After Initial Release

- Open the order that has not been changed so far and lock it from the menu: Order > Functions > Lock > Lock and save the order (IW32).
- Run the program with the same selection criteria (test run unselected).
- The program output will display the order with all required columns for case 4 (status =LKD).

Case 4b: Unlock After Locking

- Open the order that was previously locked and unlock it from the menu: Order > Functions > Lock > Unlock (IW32) and save the order.
- Run the program with the same selection criteria (test run unselected).
- The program output will display the 1 order with all required columns for case 4 (status = REL).

5. **Design Specification**
1. **Configuration**
1. **Configuration reference**

Configuration path (IMG, table, ...)	Plant Maintenance and Customer Service > Maintenance and Service Processing > Maintenance and Service Orders > General Data > User Status for Orders > Define Status Profile > Define User Status Profile for Orders

A small configuration is required for this interface to work as designed. It will need a new user status to be assigned to the current user status profile ZPM_OR1 with the following values:

- ID: CMXS.
- Description: Sent to CMX.

1. **Purpose of configuration**

a. The configuration above in reference to a SAP Note required to activate change documents for classification for functional locations.

2. Workflow

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

N/A

Object Name	Object Type	Object Description

2. Flow Diagram

<<< Please provide the technical process flow diagram >>>

N/A

3. Steps Description

<<< Process steps should be descriptive in nature. The aim of the process step is to describe the overall technical process >>>

N/A

4. Technical Details

N/A

Trigger Mechanism	Mention the start condition for the workflow, e.g. on creation of a purchase document, batch program etc.
Start Condition	Example – The workflow should start only for certain document type, workflow should start only if credit amount is greater than 250000 etc.
Business Object	Mention the business object, if possible. Otherwise indicate the object in general terms (e.g., Purchase Requisition)
Standard Workflow Task / Template	In case of enhancement required for delivery workflow is required.
Level of Approval Required	

Agent Determination Technique	Role - Security Role Org Unit - HR Org Structure Custom Table - Agents in custom table Distribution Lists Unspecified - To be decided in Functional Specification Other <use to elaborate on a selection of "Other"> If the agent determination technique is different for each foreground step then please repeat this section.
Mention Logic for Agent Determination (if any)	
Notification Destination	Internal User (Mail Inbox) External User (email address)
Workflow Notifications Text	If any specific work item text/work item subject to be used.
Escalation Handling (if any)	If any deadline monitoring is to be done. Example: If approver does not approve for 3 business days notify his supervisor.
Integration with Portal	
Configuration Dependencies	Example setting up a new organization structure.
Error Handling (if any)	An exception situation could occur if workflow routes to a one position is vacant/not available (i.e. no user is assigned to that position.) If a specific report or additional information is required. Add attachment if necessary.
Substitution	

5. Authorization

<<< Explain which roles should be added or used to approve/reject and execute workflow items. Enter any custom authorization required >>>

N/A

No	Business Catalog	Authorization Parameter	Parameter Value

3. **Report**

1. **Technical Reference**

<<< Technical Object References (class, program, t-code, ...) >>>

N/A

Object Name	Object Type	Object Description

2. **Selection Screen Details**

<<< The functional designer should be able to detail exactly what he/she wants at the selection screen merely by using this table. The programmer will be able to construct the screen directly from the details in this table. Some technical knowledge will be needed for the complete production of this table>>>

N/A

Name	Type	Parameter or Select Option	Comments (Range, Single/Multiples selections, Patterns Mandatory etc.)	Default Value
	Table-Field Check Box Radio Button with Group	Parameter Select Option		
	Table-Field Check Box Radio Button with Group	Parameter Select Option		

3. **Desired Screen Design**

<<< Enter attachment if necessary >>>

N/A

4. **Technical Details**

<<< Information like relevant database tables, data retrieval logic, type of report like (simple list report or ALV), sorting order, detail functionality, other display attributes, special interaction on clicking one or more columns etc. can mentioned here >>>

N/A

5. **Starting Conditions**

<<< When should the report be run? Does an interface need to be run before the report is valid, and (more commonly), should it be a batch only program (with added security) or is it needed on-line as well?

E.g. 'This program will be run after month-end billing.

E.g. 'This program will be run each time a sales order is saved >>>

N/A

6. **Data Mapping Tables**

<<< List of all the fields along with their details are to be mentioned here. Look and feel wise, a desired report design can also be specified here >>>

N/A

Field Name	Field Description	Output Length	Output Type	Format	Position	Screen No / Field Name

7. **Report Example**

<<< Use Attachment if necessary >>>

N/A

8. **Authorization**

<<< Explain which roles should be added or used for these reports. Enter any custom authorization if required >>>

N/A

No	Business Catalog	Authorization Parameter	Parameter Value

4. **Interface**

1. **Technical Reference**

<<< Technical Object References (class, program, t-code, ...) >>>

Object Name	Object Type	Object Description

2. **Technical Details**

Interface Name	
Direction (with respect to this system)	Inbound Outbound other If other, please specify exactly
Interface Type	Batch near real-time real-time other If other, please specify exactly
Interface Frequency	Hourly Details: Daily Details: Weekly Details: Monthly Details: Quarterly Details: Yearly Details: On-Demand Details: Other Details:
Type of Records Sent	Delta Fields Delta full-record other If other, please specify exactly
Volume (per single execution)	Average Volume: <Volume> records per interface execution Peak Volume: <Lower Volume – Upper Volume>

3. **Flow logic**

<<< Please explain any flow logic, calculations, rules, etc.. that should be implemented in this interface >>>

4. **Interface Data Layout**

<<< Please list the source and destination data elements, plus any mapping that will be required for this interface. If IDOC, include segment name in structure column. Excel matching this format can be attached in place of this table. >>>

Source Structure	Source Field	Description	Data Type	Length	Transformation	Target Structure	Target Field	Description	Data Type	Length	Mand / Opt.	Comments/Remarks

5. **Mapping Rules & Conversion Criteria**

<<< This section should contain any additional mapping rules and conversion criteria not covered in the previous section. >>>

6. **Special Case: Bi-Directional Real-Time Interface**

<<< If you know this interface will be a bi-directional real-time interface (i.e. the "Source" system sends and receives data in the same execution), then a second data mapping is required. If applicable, duplicate the table from Section 4.4.4 and capture the "return data" mapping rules for the "Source" system >>>

7. **Sample Data**

<<< Please provide two attachments of sample source data with the expected target data after this interface is executed. Please supply the sample data in the native format or .csv, and preferably zipped >>>

8. **Data Retention**

<<< In file based interfaces a "backup" copy of interface data can be retained in the middleware for each execution. This can be useful for reconciliation purposes. Please indicate the retention period for this interface. If not file based, then the source or target system must fill any data retention requirements >>>

Selection		Comments
	None	
	7 Days	
	15 Days	
	30 Days	
	Other	

9. Middleware Solution

<<< This section should contain an outline of the chosen middleware solution and the processes involved. Middleware specific configuration should be specified >>>

10. Interface Scheduling

<<< Please describe any requirements around the timing of this interface >>>

11. Authorization

<<< Explain which roles should be created / added or users / IT for reprocessing errors or ad hoc requests. Enter any custom authorization if required. Enter the file path or folder structure to which users/IT will need access to >>>

No	Business Catalog	Authorization Parameter	Parameter Value

12. Other system documentation

<<< Reference the other system's documentation, when relevant >>>

5. Conversion

1. **Technical Reference**

<<< Technical Object References (class, program, t-code, ...) >>>

N/A

Object Name	Object Type	Object Description

2. **Technical Details**

N/A

Conversion Name	

3. **Conversion Data Layout**

<<< Please list the source and destination data elements, plus any mapping that will be required for this conversion. If uploading from file, source structure can be omitted. Excel matching this format can be attached in place of this table. >>>

N/A

Source Structure	Source Field	Description	Data Type	Length	Transformation	Target Structure	Target Field	Description	Data Type	Length	Mand/Opt.	Comments/Remarks

4. **Mapping Rules & Conversion Criteria**

<<< This section should contain any additional mapping rules and conversion criteria not covered in the previous section. >>>

N/A

5. **Sample Data**

<<< Please provide two attachments of sample source data with the expected target data after this conversion is executed. Please supply the sample data in the native format or .csv, and preferably zipped >>>

N/A

6. **Authorization**

<<< Explain which roles should be created/added or used for loading data. Enter any custom authorization if required >>>

N/A

No	Business Catalog	Authorization Parameter	Parameter Value

6. **Enhancement**

1. **Business Add-Ins (BADIs)**

N/A

BADI Property	Value/Object
System	*<<< BTP, S/4 HANA, ...>>>*
Transaction	
Enhancement Spot	
BADI Name	
Enhancement Implementation	
BADI Implementation	
Class	
Method	
Filter	
OData Service	

2. **Implicit Enhancement**

N/A

Property	Value/Object
Transaction	
Enhanced Object	
Implementation	

3. **User-Exits**

N/A

Property	Value/Object
Transaction	
Main Program	
Includes	
Form Routines	

4. **CDS Views Extension**

1. **Technical Reference**

N/A

Property	Value/Object
Original CDS View Name	
Extended CDS View Name	
Purpose of Extension	
Extension Type	☐CDS View Extension ☐Custom CDS consuming Standard CDS ☐View with Additional Associations or Joins ☐Metadata Extension
Odata Exposure	☐Yes ☐No
Input Field Parameters	
Service Definition	
Service Binding	

2. **Fields Added**

N/A

Field Name	Data Element	Source Table	Description	Annotations

5. **Function Exits**

N/A

Property	Value/Object
Transaction	
Enhancement	
Function Module Name	
Includes	

6. **Field Exits**

N/A

Property	Value/Object
Enhancement	
Main Program Name	
Function Module Name	
Field Exit Id	
Screen Number	
Screen Field Name	
Conditions for execution	

7. **Menu Exits**

N/A

Property	Value/Object
Enhancement	
Menu/Path	
Function/Transaction Code	

8. **Screen Exits**

N/A

Property	Value/Object
Enhancement	
Main Program Name	
Screen Number	
Program Name & Sub-Screen Number	

9. **Search Help Exits**

N/A

Field Name	Field Description	Import / Export (I/E)	Key Field (Y/N)	Data Element	Type (CHAR, NUMC)	Length	Default Value

10. **Search Help assignment**

N/A

Property	Value/Object
Standard Search Help	
Collective Search Help	
Elementary Search Help	

11. **Business Transaction Events (BTE)**

N/A

Property	Value/Object
Transaction	
BTE Number	
Product Name	

Function Module	

12. Custom Transaction

<<< Functional details of custom transaction can be incorporated here. Number of screens required and flow diagram can be included and provide the selection screen shot along with the table name and field name and screen shot for the required output >>>

N/A

13. Requirement routine

N/A

Menu/Submenu	
Routine number	
Business logic required	

14. Substitution

N/A

Validation Description	Fields required for validation	Point of Validation	Table used in validation	Business Rules

Substituted Field	Derived from Field	Table used in Substitution	Business Rules

15. Flow logic

<<< Please explain any flow logic, calculations, rules, etc that should be implemented in this enhancement >>>

N/A

16. Authorization

<<< Which authorization object should be used for controlled execution? Enter any custom authorization if required >>>

N/A

No	Business Catalog	Authorization Parameter	Parameter Value

7. Form

1. Technical Reference

<<< Technical Object References (class, program, t-code, ...) >>>

N/A

Object Name	Object Type	Object Description

2. Form Layout

<<< Refer to the following for an output samples for Window mapping, Label Description and Field mapping >>>

N/A

C:\Documents and Settings\sutapa\My D C:\Documents and Settings\sutapa\My D

3. Layout Windows

N/A

Reference	Print on page	Label Position
		X : Y :
		X :

		Y :
		X : Y :
		X : Y :
		X : Y :
		X : Y :
		X : Y :

4. **Field Mapping**

N/A

Field	Field Description	Functionality	Logic	Print on page	Font	Font Format	Window

5. **Standard Texts / Text Modules**

N/A

Reference	Text	Print on page	Label Position	Font	Output Format	Font Format

6. **Translation**

N/A

Reference	Description of use (in Language1)	Description of use (in Language2)	Description of use (in Language3)	Text Module Name	Notes

7. **Layout Details**

N/A

Position of Left Margin (specify unit)	
Position of Right Margin (specify unit)	
Position of Logo (specify unit)	
Logo (specify logo)	
Position of Main Window (specify unit)	

8. **Flow logic**

<<< Please explain any flow logic, calculations, rules, etc that should be implemented in this form >>>

N/A

9. **Authorization**

<<< Explain which roles should be created/added or used for printing and testing forms. Enter any custom authorization if required >>>

N/A

No	Business Catalog	Authorization Parameter	Parameter Value

8. **Fiori Application**

1. **Header Information**

N/A

Application Title	
Application ID	
Type of Enhancement	☐Custom Application ☐ Standard Application
Development Type	*<<<Fiori Elements App ....Free Style UI5 App>>>*
Application Type	*<<<List Report, Object Page , Over view Page ,etc >>>*
UI Enhancements	☐Custom Fields Added ☐UI Layout Modified ☐Extensibility Hook Used ☐Fragments or Views Introduced

2. **Technical Reference**

N/A

Object Name	Object Type	Object Description
<<< Odata Object >>>		
<<< CDS View >>>		
<<< Custom Fields >>>		
<<< Catalogs >>>		
<<< Rules >>>		

3. **Desired Screen Design**

<<< Enter attachment if necessary >>>

N/A

4. **Technical Details**

<<< Information like relevant database tables,CDS Views,ODATA services , data retrieval logic, detail functionality, other display attributes, special interaction on clicking one or more columns etc. can mentioned here >>>

N/A

5. **Authorization**

<<< Enter Authorization Objects/fields, to be used and specific user Groups >>>

N/A

No	Business Catalog	Name of Space (L2)	Name of Page (L3)	Name of Section (L4)	Name of App/Tile(L5)	Authorization Parameter	Parameter Value

6. **Custom Tables/Structure**

<<< This section should detail the attributes of any new custom table created for one of the above sections, and the properties of its fields.

NB: Existing Data Elements and/or Domains should be used whenever possible when creating custom table fields, in order to avoid unnecessary typos. In this instance, the data table row for that field should not be completed beyond 'Domain', as the remaining attributes will be default values for the selected Domain. >>>

N/A

Table Name	
Short text	
Size category	
Table maintenance allowed	
Maintenance Type	Manual / Automatic Maintenance (application table) Transportable Maintenance (customizing table)
Data class	
Buffering	
Table maintenance generation	
Authorization Group	
Change Log Enabled (Y/N)	

(mandatory for GxP related table)								
SPRO Path *(mandatory for customizing tables)*								
Field Name	**Data Element**	**Domain**	**Type**	**Length**	**Check Table-Field**	**Key Field**	**Foreign Key**	**Description**
Comments								

7. **Error Handling**

<<< Provide Error Handling details here. Job run notifications, error notifications, E-Mail messaging, custom programming, etc. may be required >>>

1. **Error Messages**

<<< Describe the expected error messages for different error conditions >>>

Note: Error Messages S = SAP Origin, I = Integration Origin, C = CMX Origin

Error Message Number	**Error Message Text (70 characters)**	**Error Conditions**
01S	The message must be displayed exactly as the BAPI has returned it	When BAPI_ALM_ORDER_MAINTAIN returns an error message

8. Validation

1. Test Case References

<<< List the Test Case(s) used to validate the functionality / configuration covered in this document (IQ / OQ). >>>

Test Case ID	Test Case	Comment